Homework 5 — LoRA Fine-Tuning of SmolLM-135M

Advanced AI and Machine Learning
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1 Setup and Experimental Configuration

This report describes a supervised fine-tuning (SFT) experiment on `HuggingFaceTB/SmolLM-135M` using Low-Rank Adaptation (LoRA), together with an ablation study on the LoRA rank r. Training was run on a Google Colab T4 GPU using the Hugging Face `trl` and `peft` libraries.

Component	**Value**
Base model	`HuggingFaceTB/SmolLM-135M`
Dataset	`maxmyn/wholesome_greentext_110k` (first 100 rows)
Target modules	`q,k,v,o,up,down,gate_proj`
LoRA rank r (main run)	16
LoRA rank r (ablation)	4
LoRA α	32
LoRA dropout	0.05
Learning rate	2×10^{-4}
Batch size per device	4
Max training steps	500
Mixed precision	fp16
Sampling (inference)	$T = 0.7,$ top-$p = 0.9$

Table 1: Hyperparameter summary for the two training runs.

Perplexity is computed as $\text{PPL} = \exp(\mathcal{L}_{\text{CE}})$, where $\mathcal{L}_{\text{CE}}$ is the average cross-entropy loss. A critical caveat that recurs in the analysis below is that *evaluation is performed on the training set itself* (`eval_dataset=train_dataset` in the notebook), so the reported perplexities measure fit, not generalisation.

2 Training Loss (Weights & Biases)

Figure 1 shows the training loss logged to W&B for the main run ($r = 16$).

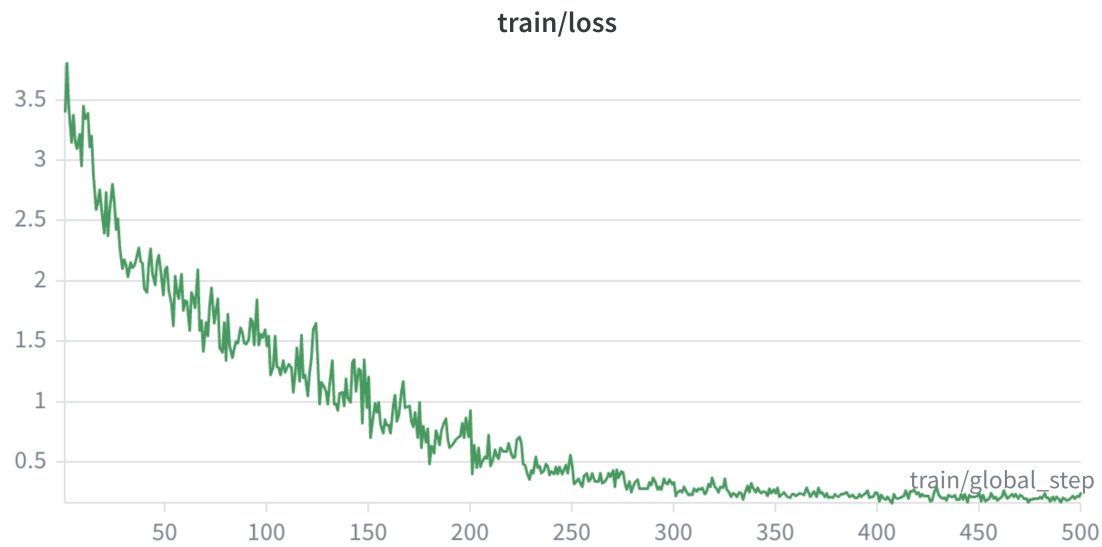

Figure 1: Training loss vs. global step, LoRA $r = 16$. Logged every step; W&B applies light smoothing at this zoom level.

Observations.

- Loss falls from ≈ 3.8 at step 0 to ≈ 0.2 by step 500, roughly a $19\times$ reduction.
- Three phases are visible:
  - **Steps 0–100:** rapid descent from ≈ 3.8 to ≈ 1.5. The model is learning the *surface format* of greentexts — the > prefix, line breaks, and short imperative clauses.
  - **Steps 100–250:** steadier decline to ≈ 0.5 as content patterns start being fit.
  - **Steps 300–500:** plateau around 0.2–0.3.
- The plateau at such a low loss is a strong indicator of **memorisation rather than generalisation**: with 100 examples, batch size 4, and 500 steps, each example is seen approximately 20 times. This is an unusually high number of passes over a tiny set, so a near-zero loss is expected but not especially informative.
- High per-step noise throughout is consistent with small-batch training (batch $= 4$): each step's gradient is dominated by the few sequences in it.

3 Perplexity Across Checkpoints

Perplexity was evaluated at steps $\{0, 100, 200, 300, 400, 500\}$ for both the main run ($r = 16$) and the ablation run ($r = 4$). Values are reproduced from the notebook outputs.

Step	**PPL** (r=16)	**PPL** (r=4)
0	29.96	29.95
100	4.68	4.37
200	1.49	2.55
300	1.25	1.44
400	1.19	1.29
500	1.28	1.29

Table 2: Perplexity at checkpoint steps for both LoRA configurations.

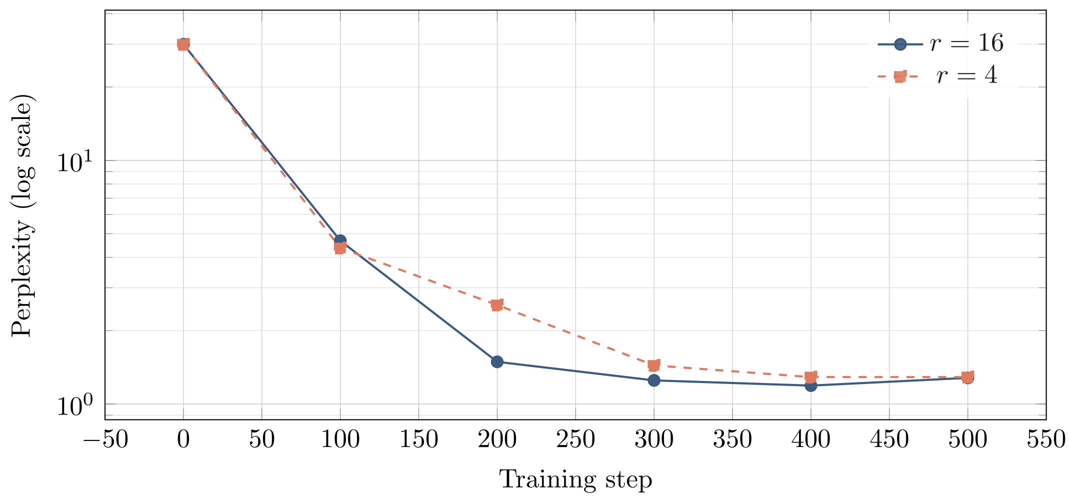

Figure 2: Perplexity across training checkpoints for the two LoRA ranks. Both configurations start from the same pre-trained base and converge to essentially the same value; $r = 16$ reaches low perplexity faster.

Observations.

- Both runs start at ≈ 30 (identical pre-trained weights) and converge to ≈ 1.2–1.3 by step 500. The final perplexities are effectively indistinguishable.
- r=16 reaches a low perplexity *faster* than r=4 (1.49 at step 200 vs. 2.55 for r=4). More trainable parameters $\Rightarrow$ more capacity to fit the small training set quickly. This is the expected behaviour.
- A very slight uptick appears at step 500 for r=16 ($1.19 \rightarrow 1.28$). On such a tiny dataset this is within noise, but it is also the region where further training no longer helps.
- **Methodological caveat.** Because the evaluation set *is* the training set, a perplexity of ≈ 1.2 reflects memorisation, not language modelling quality on unseen text. With a proper held-out split, held-out perplexity would be noticeably higher. The shapes of the two curves are still informative as a relative comparison, but the absolute numbers should not be read as a measure of model quality.

4 Text Generation: Before vs. After Fine-Tuning

Four prompts adapted from the dataset structure were passed to the base model and to the fine-tuned models. Representative generations are reproduced below (truncated for readability). Inference used $T = 0.7$, top-$p = 0.9$.

Prompt 1: Treasure hunt

```
> Embark on a global treasure hunt
> Decode cryptic clues
>
```

Base model (before fine-tuning).

```
> Become a legendary treasure hunter
> Become the fastest and strongest pirate in the sea
> Outsmart rival pirates and become a legendary sea creature
> Embark on a quest to find hidden treasure
> Embark on a global treasure hunt
> Embark on a quest to find hidden treasure
> Embark on a global treasure hunt
> Embark on a quest to find a Goldentimedelta
```

Fine-tuned, $r = 16$.

```
> Become a legendary treasure hunter
> Become the sought-after master thief
> Embark on a life changing adventure
> Embark on a global treasure hunt
> Embark on a quest to become the most feared treasure hunter
> Embark on a journey to find the treasure
> Embark on a quest to become the most feared treasure hunter
```

Fine-tuned, $r = 4$ (ablation).

```
> Become a legendary treasure hunter
> Become the captain of a team
> Claim the highest prize in the game
> Embark on an epic quest
> Become a legendary treasure hunter
> Become the captain of a team
> Claim the highest prize in the game
```

Prompt 2: Cashier

```
> Work as a cashier
> Customer counts their change
> Realize the importance of patience
>
```

Base model (before fine-tuning).

```
> Customer appreciate their mistakes
> Accept their portion
> Order food from a vendor
> Live a life of gratitude
> Realize the true meaning of gratitude
> Eat more food of appreciation
> Live a life of gratitude
```

```
> Eat more food of appreciation
```

Fine-tuned, $r = 16$.

```
> People might buy more than they thought
> Sometimes, it's better to save a portion of your earnings
> Eat chips
> Eat a burrito
> Eat a bag of chips
> Get a job as a waitress
> Realize the value of patience
> Money is precious
```

Fine-tuned, $r = 4$ (ablation).

```
> Customer often forget to pay
> Order extra change and encourage
> Customers to chip in
> Order more
> Happy customer and newfound appreciation for patience
> Order more
> Order more
> Order more
> Order more
> Order more ...  (repeats 13 times)
```

Prompt 3: Pompeii

```
> Explore the ruins of Pompeii
> Uncover ancient artifacts
>
```

Base model (before fine-tuning).

```
> Become an expert on Pompeii relics
> Share your knowledge with others
> Become a renowned archaeologist
>>>>>>>>>>>>>:>:>>>:>:>:>:>>>>>>>>>>>>>>>>>>>>>
```

Fine-tuned, $r = 16$.

```
> Become a numismatic historian
>
> By Jack
```

Fine-tuned, $r = 4$ (ablation).

```
> Meet an archaeologist
> Collaborate with them
> Become a respected archaeologist
> Challenge traditional roles
> Realize the importance of collaboration
> +>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>
```

Prompt 4: Homeless person

```
> Take on the role of a homeless person
> Experience the hardships and stigma
> Learn the true meaning of resilience
>
```

Base model (before fine-tuning).

```
> Learn the true meaning of resilience
> Realize the strength of your spirit
> Embrace the path of redemption
> Live a life of hope and fulfillment
> Eat tacos and laugh until tears come
> Accept the loss of innocence
> Live as a documentarian
```

Fine-tuned, $r = 16$.

```
> Realize the strength of empathy and kindness
> Practice putting yourself in someone else's shoes
> Appreciate the small acts of kindness
> Realize the power of empathy
> Accept the burden of leaving your own life behind
> Live a life of gratitude and generosity
> Embark on a journey of self-discovery and fulfillment
```

Fine-tuned, $r = 4$ (ablation).

```
> Better understand the struggles of the homeless
> Appreciate the small acts of kindness
> Challenge societal norms to give homeless people a second chance
> Find empathy for the ones in need
> Realize the power of kindness
> Join the movement for destigmatization
> Advocate for the rights of the homeless
```

5 Discussion of Generation Quality

Format acquisition (positive). The base model already produces a `>`-prefixed structure because the prompt primes it, but it breaks down at length — Prompt 3 degenerates into `>>>>>>>:>:>>>`, which is not valid greentext. After fine-tuning, both r=16 and r=4 produce a consistent structure across all four prompts: short imperative clauses, one per line, each starting with `>`. This corresponds directly to the first phase of the loss curve (steps 0–100), during which the model rapidly learns surface format.

Semantic coherence (mixed). Fine-tuning improves topical grounding. For the homeless-person prompt, the fine-tuned r=16 model stays on theme ("empathy", "kindness", "putting yourself in someone else's shoes"). For the cashier prompt it produces at least two coherent continuations before drifting toward food items. However, coherence is shallow: the model frequently switches topic or inserts lines that do not fit the running narrative (e.g. "Eat chips" in the cashier continuation for r=16).

Repetition as the dominant failure mode. This is the single clearest symptom in the generations and the most important remark. The r=4 model produces "Order more" thirteen times in a row for the cashier prompt; r=16 loops "Embark on a quest to become the most feared treasure hunter" for the treasure prompt. This is a textbook signature of **overfitting on a tiny dataset combined with many effective epochs**: the model has learned that a handful of phrases from the training set are high probability and collapses onto them at sampling time. Inference-time mitigations such as a repetition penalty or lower sampling temperature would mask the symptom; the underlying fix is more training data, a held-out validation split, and early stopping.

Qualitative r=16 vs. r=4. r=16 occasionally introduces unusual vocabulary ("numismatic historian", "documentarian"), while r=4 outputs stay closer to training-set n-grams and repeat more. This aligns with the theory: lower rank $\Rightarrow$ fewer trainable parameters $\Rightarrow$ less representational capacity for variation, so the model leans harder on memorised phrases. On this dataset the difference is noticeable but not large.

Robustness gaps. Even after fine-tuning, the Pompeii prompt causes early degeneration for both ranks (`> By Jack` for r=16; a `+>>>...` tail for r=4). With only 100 training examples, it is unrealistic to expect coverage of arbitrary topics.

6 Discussion of Hyperparameter Impact

- **LoRA rank r (16 vs. 4).** The main ablation. Higher rank trained faster (lower perplexity at steps 100–200) and produced marginally more varied outputs. Final perplexities are essentially identical. Trade-off: r=4 has roughly 4× fewer trainable LoRA parameters and is cheaper to train and serve. On a 100-example dataset the capacity difference does not justify r=16; on a 10^4+ example dataset a meaningful quality gap would likely emerge.
- **α/r scaling.** I kept α=32 fixed while reducing r from 16 to 4, so α/r went from 2 to 8. This means the r=4 run effectively had a 4× larger update magnitude per LoRA direction. A cleaner ablation would scale α with r (e.g. $r=4, \alpha=8$) so the effective learning rate on the LoRA branch is constant across runs. This is a limitation of the ablation design.
- **Learning rate 2×10^{-4}.** Standard for LoRA fine-tuning. The smooth loss descent without divergence suggests it is reasonable; I did not sweep alternatives, so this should be treated as a default rather than a tuned value.
- **Dataset size (100 examples).** The dominant factor in this experiment. 100 examples × 500 steps ÷ batch 4 means each sequence is seen ≈ 20 times. The near-zero training loss, the ≈ 1.2 training-set perplexity, and the repetition observed in generation are all consequences of this. Any conclusions about the other hyperparameters are conditional on this severe data limitation.
- **fp16.** Correct choice on T4 (no bf16 support); noted for reproducibility.
- **Sampling hyperparameters (T = 0.7, top-p = 0.9).** These affect output quality independently of training. Adding `repetition_penalty=1.2` or lowering T would reduce visible repetition without retraining, but would not address the underlying memorisation.

7 Limitations and What I Would Do Differently

- **Train/eval split.** Perplexity was evaluated on the training set. A 10–20% held-out split would give a meaningful generalisation estimate.
- **Dataset size.** 100 examples is far below what is needed for robust generalisation. Scaling to 10^3–10^4 examples would shift the plateau point and visibly reduce generation repetition.
- **Ablation design.** Varying r alone while fixing α confounds rank with effective update magnitude. A future run should sweep (r, α) pairs that keep α/r constant.
- **Prompt coverage.** Only four prompts were used for qualitative evaluation; a larger, disjoint prompt set would give a fairer before/after comparison.
- **Quantitative generation metric.** No numerical generation metric was computed. Adding `distinct-n` would numerically confirm the repetition observation; BLEU or ROUGE against held-out references would give a coarse quality score.

8 Summary

Fine-tuning SmolLM-135M with LoRA on 100 greentext examples produced a model that reliably reproduces the surface format of greentexts and shows improved topical grounding relative to the base model, but also exhibits strong repetition, characteristic of overfitting on a very small training set. An r=16 vs. r=4 ablation showed that higher rank reaches low training-set perplexity faster, but the two configurations converge to essentially the same final perplexity; on a dataset this small, r=4 is adequate. The most interesting takeaway is not the loss curve but the connection between it and the qualitative generation failures: low loss here is evidence of memorisation, and the repetitive outputs are its visible consequence.